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Artificially Intelligent People’s presentation

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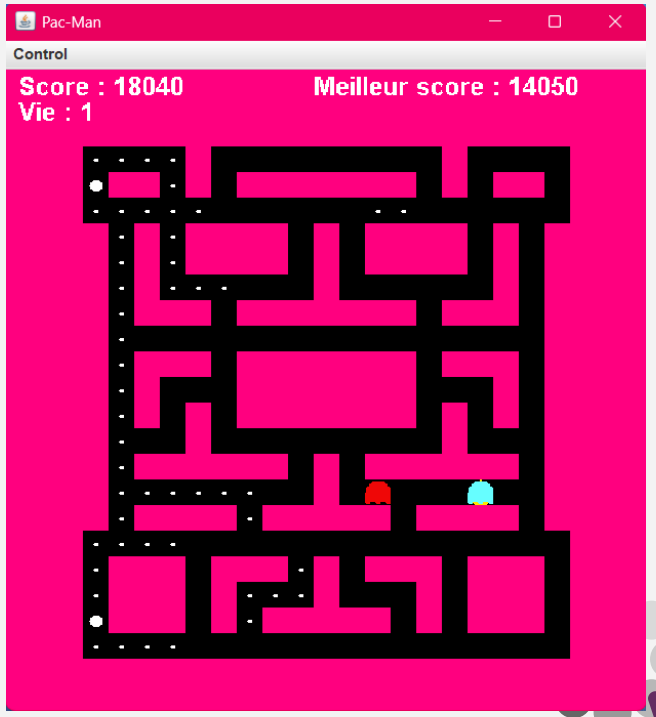
Some statistics (over 25 tests)

	Average	Median	Minimum	Maximum
Score	4073	2700	270	18040
Time (ms)	32,31	29,10	16,55	56,35
Number of actions	825	464	30	3542

< 2000 pts	8
>= 2000 pts	17
Total	25



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findNextMove()

- Get the current state of the game,
- Analyze the environment,
- Run simulations to choose the best direction by assigning scores

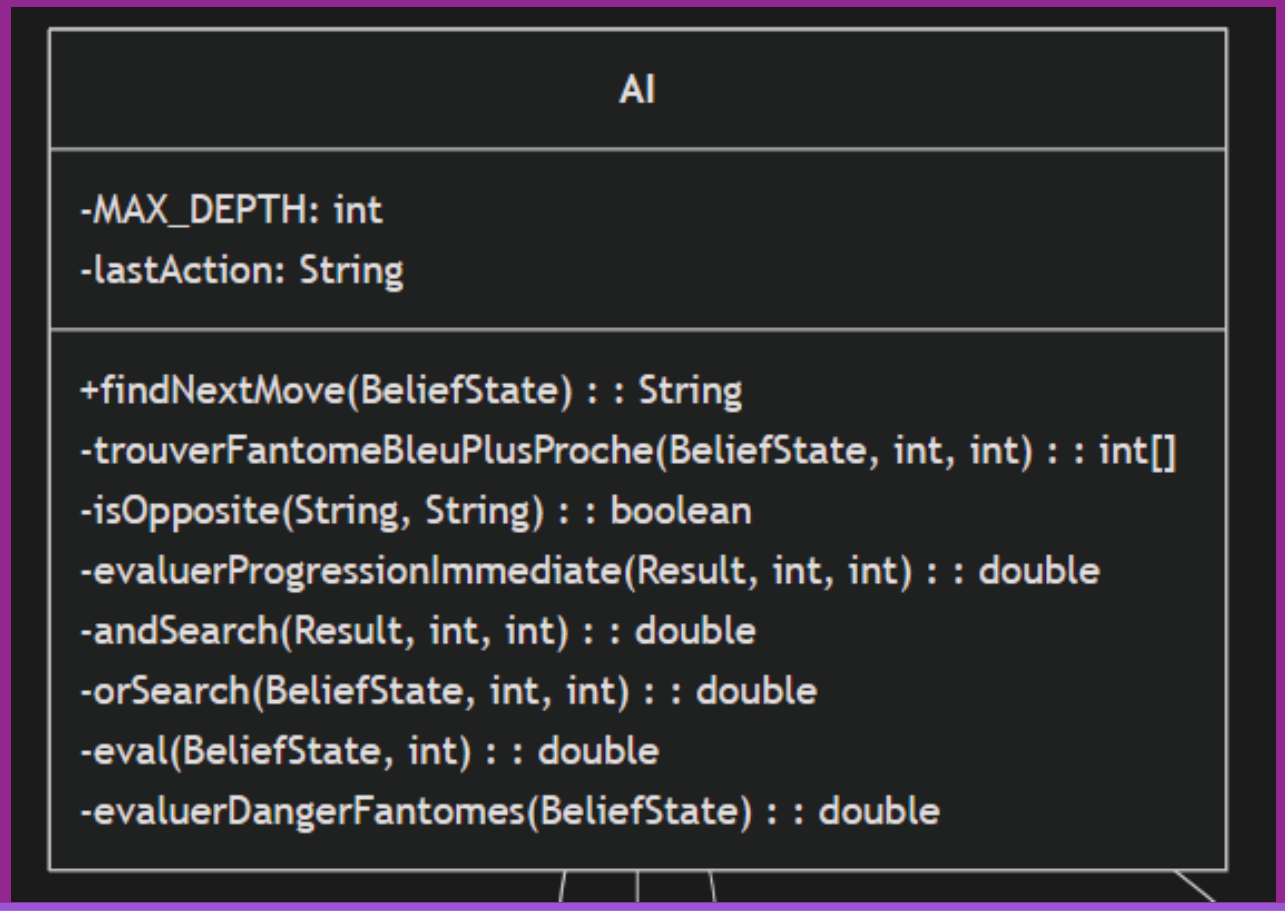
3 main phases:

1. Immediate Context analysis

2. AND-OR Search (depth 3)

3. Heuristic Adjustments

General Structure of the AI Class



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Phase 1: Context Analysis & Threat Detection

- Minimum distance to nearest gum
- Identify vulnerable ghost and nearest normal ghost
- Game state metrics :
  - Current gum count
  - Pac-Man position tracking

Ghost type differentiation :

- Scared ghosts are targets with bonus scoring
- Normal ghosts are threats requiring distance management

Special handling for ghosts at distance 1

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Phase 2: AND-OR search implementation

```
findNextMove()
     ↓
andSearch()   → Averages across
     ↓          possible belief
                states
orSearch()    → Selects best
     ↓          action branch
eval()        → Terminal state
                evaluation
                (depth 0 or game end)
```

Termination Conditions:

- Depth limit (=3) reached
- No gum remaining
- Pac-Man life ≤ 0

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OR SEARCH idea :

- Choice among multiple possible actions
- Returns best value
- Like Max in Minimax
  - PacMan is a rational player, so **he chooses an action that gives the best outcome**

AND-OR search

```java
private static double orSearch(BeliefState state, int depth, int gommesInitiales) {
    // conditions d'arret
    if (depth == 0 || state.getNbrOfGommes() == 0 || state.getLife() <= 0) {
        return eval(state, gommesInitiales);
    }

    Plans plans = state.extendsBeliefState();
    if (plans.size() == 0) return eval(state, gommesInitiales);

    double maxVal = Double.NEGATIVE_INFINITY;

    // recherche de la meilleure action
    for (int i = 0; i < plans.size(); i++) {
        Result res = plans.getResult(i);
        double val = andSearch(res, depth - 1, gommesInitiales);
        if (val > maxVal) {
            maxVal = val;
        }
    }

    return (maxVal == Double.NEGATIVE_INFINITY) ? -100000.0 : maxVal;
}
```

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AND-OR search

AND SEARCH idea :

- Average over all possible states
- Manages uncertainty
- Similar to expectation in expectiminimax
  - The game is unpredictable, when PacMan moves, sevaral things can happen
    - Since we don't know exactly what will happen, we take the average of all the possibilities.

```java
private static double andSearch(Result result, int depth, int gommesInitiales) {
    if (result.size() == 0) return Double.NEGATIVE_INFINITY;

    double totalScore = 0;
    int countValid = 0;

    for (BeliefState bs : result.getBeliefStates()) {
        double val = orSearch(bs, depth, gommesInitiales);

        // traitement different selon la valeur
        if (val > -50000) {
            totalScore += val;
            countValid++;
        } else {
            totalScore += val * 0.3;
            countValid++;
        }
    }

    return countValid > 0 ? totalScore / countValid : Double.NEGATIVE_INFINITY;
}
```

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Phase 2: AND-OR search implementation

- OR-SEARCH uses MAX because Pac-Man chooses the best possible action.
- AND-SEARCH uses AVERAGE because we must evaluate the expected results considering the game's uncertainty.

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Phase 3 :

1st heuristic : Gluttony

- +500 points for each gum eaten compared to the start of the turn. This makes him 'greedy'
- a bit of pressure: -100 points for each gum left on the map

Heuristic Adjustments & Decision Making

eval(…)

```
value = score
      + (gum_eaten * 500)
      - (remaining_gum * 100)
      - (gum_distance * weight)
      - ghost_danger
```

```
// bonus pour les gommes deja mangees
int gommesMangees = gommesInitiales - nbGommes;
score += gommesMangees * 500;

// penalite pour les gommes restantes
score -= nbGommes * 100;
```

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Phase 3 :

2nd heuristic : Dynamic Proximity

- penalize the distance from the nearest gum
- a dynamic weight :
  - 50 + (1500 / number of gums remaining)

Heuristic Adjustments & Decision Making

eval(…)

```
value = score
      + (gum_eaten * 500)
      - (remaining_gum * 100)
      - (gum_distance * weight)
      - ghost_danger
```

```
// calcul du poids de la distance
int poidsDistance = 50 + (1500 / (nbGommes + 1));

// penalite pour la distance a la gomme la plus proche
if (distGomme < Integer.MAX_VALUE) {
    score -= distGomme * poidsDistance;
}
```

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Phase 3 :

3rd heuristic : Security / Survival

- Bonus > fear ? Take some risk !
- Otherwise, escape

The evaluation corresponds to a weighted sum between the desire to eat, the urgency to finish the level and the fear of dying.

Heuristic Adjustments & Decision Making

eval(…)

```
value = score
      + (gum_eaten * 500)
      - (remaining_gum * 100)
      - (gum_distance * weight)
      - ghost_danger
```

```
// soustraction du danger des fantomes
score -= evaluerDangerFantomes(state);

return score;
```

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Pac-Man vs. the Ghosts

In findNextMove

Blue ghost proximity bonuses and...

- A penalty of -2000 will be applied if PacMan move away from the scared ghost.
- **Special case :** if a scared ghost is exactly at **distance 1** from PacMan (no matter where the other ghost is located), we add a **bonus of +3000** to force him to pursue him and eat him for the next action.

When dangerous ghosts are further away

Distance	Bonus
0	+8000
1	+5000
2	+3000

In evaluerDangerFantomes

Normal ghost danger penalties and...

- Special case if the 2 ghosts are at distance 1 from PacMan :
  - We reduce the danger (-200 instead of –1000) so that Pac-Man can escape and eat the gums.
    - Because he knows where the two ghosts are located, he can travel the map almost without fear.

Distance	Penalties
0	-100000
1	-1000
2	-300
3	-100
4	-10

Phase 3 : Heuristic Adjustments & Decision Making

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4th heuristic:
evaluerProgressionImmediate(Result result, int gommesInitiales, int distInitiale)

- **Gums eaten:** +4000 pts each
- **Distance improved:** +100 pts per square
- **Purpose:** Balance caution with action
- **Impact:** More aggressive, less hesitant

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Issues we encountered / Areas for enhancement

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1) Find a good heuristic
2) Find the right balance between Pac-Man's fear andhunger
3) **The problem of Pac-Man refusing to eat an isolated gum when chased by two overlapping ghosts**
4) Solve the oscillation problem
5) Solve the cycle problem
6) Solve the Pac-Man immobility problem

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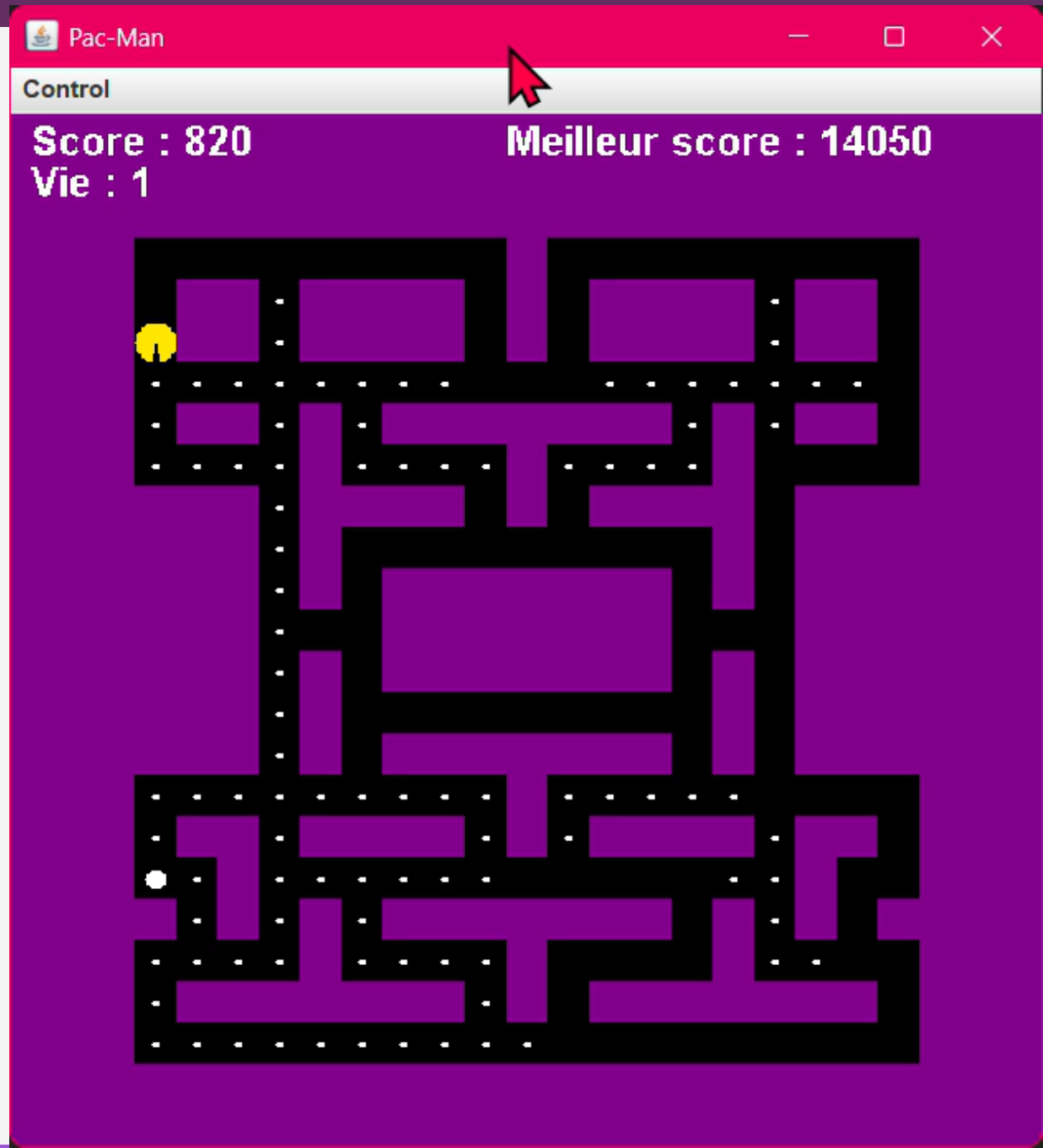
Video testing of our Pacman (fast forward)



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Solution “post-rendu” : no cycle and no oscillation

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Thank you for this project, it was fun and interesting !
Quite hard for us though…